

Abstract

COGANT (Codebase-to-GNN Translation) is a system for turning software repositories into structured Active Inference artifacts expressed in the Active Inference Institute's **Generalized Notation Notation** (GNN)¹. It occupies the space between classical program analysis—which already reasons over graphs such as call graphs and code property graphs [yamaguchi2014modeling]—and data-driven methods that expect tensors, consistent node and edge vocabularies, and exportable training bundles [allamanis2018survey; wu2020comprehensive; scarselli2009graph].

The design centers on a **program graph IR**: nodes denote entities (for example functions and variables), edges denote relations (for example calls and data flow), and each assertion carries **confidence** and **provenance** so that downstream models can treat uncertainty explicitly. A **fixpoint translation engine** applies declarative rules iteratively until no new semantic mappings emerge, resolves overlapping mappings by confidence, and reports coverage gaps explicitly. **Dynamic enrichment** from coverage reports and